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Sheaf cohomology

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Introduction

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Part I

Category theory

Chapter 1

Category Theory

Definition 1.0.1 (Category)**.** A **category** C consists of:

- A collection of *objects* $X, Y, \ldots$
- For each pair of objects X, Y a set of morphisms $\mathrm{Hom}_{\mathsf{C}}(X, Y)$. We denote this morphisms by $f : X \to Y$ or $X \xrightarrow{f} Y$.

such that

- Each object has an *identity morphism* $\mathbb{1}_X : X \to X$.
- For each pair of morphisms $X \xrightarrow{f} Y \xrightarrow{g} Z$ there exists a *composite morphism* $gf : X \to Z$.

We impose two axioms on this data

1. For any $f : X \to Y$ the composites $\mathbb{1}_Y f$ and $f\mathbb{1}_X$ are equal to f.
2. For any triplet of morphisms $X \xrightarrow{f} Y \xrightarrow{g} W \xrightarrow{h} Z$ the composites $h(gf)$ and $(hg)f$ are equal. We denote this composite by hgf.

⌟

Remark 1.0.1. Technically speaking Definition 1.0.1 is the definition of a *locally small category*. In general, a category C can have an arbitrarily *large* collection of morphisms between two objects. ▽

Every Category has an associated opposite category

Definition 1.0.2 (Opposite Category)**.** Let C be a Category. The **opposite Category** of C, C^{op} has

- The same objects as C.
- For every pair of objects $X, Y \in \mathsf{C}^{\mathrm{op}}$: $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(X, Y) = \mathrm{Hom}_{\mathsf{C}}(Y, X)$.

⌟

Definition 1.0.3. A morphism $f : X \to Y$ in a Category C is **monomorphism** if for any parallel pair of morphisms $h, k : W \rightrightarrows X, fh = fk$ implies $h = k$. Dually, f is an **epimorphism** if for any pair of morphisms $h, k : Y \rightrightarrows Z, hf = kf$ implies $h = k$. ⌟

Definition 1.0.4 (Isomorphism)**.** A morphism $f : x \to y$ in a Category C is an **isomorphism** if there exists a morphism $g : y \to x$ in C such that $fg = \mathbb{1}_y$ and $gf = \mathbb{1}_x$.

The objects x and y are called *isomorphic* if there exists an isomorphism between them, and we denote this by $x \cong y$. ⌟

Notation. We denote monomorphisms by $f : X \rightarrowtail Y$, epimorphisms by $f : X \twoheadrightarrow y$ and Isomorphisms by $f : X \xrightarrow{\sim} Y$.

Definition 1.0.5 (Functor)**.** A **functor** $F : \mathsf{C} \to \mathsf{D}$ between two Categories consists of:

- For each object c in C an object $Fc \in \mathsf{D}$.
- For each morphism $f : c \to c'$ in C a morphism $Ff : Fc \to Fc'$ in D.

satisfying the following axioms

1. For each element c in C, $F\mathbb{1}_c = \mathbb{1}_{Fc}$.
2. For any composable pair of morphisms f, g in C, $Fg \cdot Ff = F(g \cdot f)$.

⌟

Notation. A Functor of the form $\mathsf{C} \to \mathsf{D}$ is usually called a *covariant functor* from C to D while a functor of the form $\mathsf{C}^{\mathrm{op}} \to \mathsf{D}$ is called a *contravariant* functor from C to D.

Example 1.0.1 (Posetal category of open sets)**.** The Functor $\mathsf{Op} : \mathsf{Top} \to \mathsf{Poset}$ sends a topological space X to its poset of open sets.

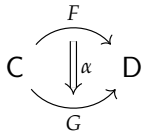
Therefore for every topological space with have an associated *posetal category of open sets* $\mathsf{Op}(X)$. ◊

Definition 1.0.6 (Natural transformation)**.** Let C, D be two Categories and $F, G : \mathsf{C} \rightrightarrows \mathsf{D}$ be two Functors. A **natural transformation** $\alpha : F \Rightarrow G$ consists of:

- For each object $c \in \mathsf{C}$ a morphism$\alpha_c : Fc \to Gc$ in D such that, for any morphism $f : c \to c'$ in C the following diagram commutes in D

$$\begin{array}{ccc} Fc & \xrightarrow{\alpha_c} & Gc \\ {\scriptstyle Ff}\downarrow & & \downarrow{\scriptstyle Gf} \\ Fc' & \xrightarrow[\alpha_{c'}]{} & Gc' \end{array}$$

We denote this by



⌟

Definition 1.0.7. A Natural transformation $\alpha : F \Rightarrow G$ is a **natural isomorphism** if every component α_f is an Isomorphism.

In this case, we say that F and G are **naturally isomorphic** and we denote it by $F \cong G$. ⌟

Definition 1.0.8. A Functor $F : \mathsf{C} \to \mathsf{D}$ is

- **full** if for all pair of objects $x, y \in \mathsf{C}$ the map $\mathrm{Hom}_{\mathsf{C}}(X, Y) \to \mathrm{Hom}_{\mathsf{D}}(Fx, Fy)$ is surjective.
- **faithful** if for all pair of objects $x, y \in \mathsf{C}$ the map $\mathrm{Hom}_{\mathsf{C}}(x, y) \to \mathrm{Hom}_{\mathsf{D}}(Fx, Fy)$ is injective.

- **essentially surjective on objects** if for every object $d \in \mathsf{D}$ there exists $c \in \mathsf{C}$ such that $d \cong Fc$.

If F is both full and faithfull we will say it is **fully faithful**. ⌟

Definition 1.0.9 (Concrete category)**.** A Category C is a **concrete Category** if there exists a faithul functor $F : \mathsf{C} \to \mathsf{Set}$. ⌟

Definition 1.0.10 (Diagram)**.** A **diagram** in a Category C with shape J is a Functor $F : \mathsf{C} \to \mathsf{J}$. ⌟

Definition 1.0.11 (Initial and terminal objects)**.** Let C be a Category. An object $i \in \mathsf{C}$ is **initial** if for every object $c \in \mathsf{C}$ there exists a unique morphism $i \to c$. Dually, an object $t \in \mathsf{C}$ is **terminal** if for every object $c \in \mathsf{C}$ there exists a unique morphism $c \to t$. ⌟

Definition 1.0.12 (Zero object)**.** An object $0 \in \mathsf{C}$ is a **zero object** if it is both an initial and terminal object. ⌟

Definition 1.0.13. Let c be an object in a Category C. The **Functor represented** by c is the Functor $\mathrm{Hom}_{\mathsf{C}}(c, -) \to \mathsf{Set}$ defined as

$$\begin{array}{ccc} x & & \mathrm{Hom}_{\mathsf{C}}(c,x) \\ {\scriptstyle f}\downarrow & \longmapsto & \downarrow{\scriptstyle f_*} \\ y & & \mathrm{Hom}_{\mathsf{C}}(c,y) \end{array}$$

⌟

Remark 1.0.2. If we consider the Opposite Category C^{op} in Definition 1.0.13 for each $c \in \mathsf{C}^{\mathrm{op}}$ we get a Functor $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(c, -) \to \mathsf{Set}$ defined as

$$\begin{array}{ccc} x & & \mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(c,x) \\ {\scriptstyle f}\downarrow & \longmapsto & \downarrow{\scriptstyle f_*} \\ y & & \mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(c,y) \end{array}$$

Therefore since $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(c, x) = \mathrm{Hom}_{\mathsf{C}}(x, c)$ we get a functor $\mathrm{Hom}_{\mathsf{C}}(-, c) \to \mathsf{Set}$

$$\begin{array}{ccc} x & & \mathrm{Hom}_{\mathsf{C}}(x,c) \\ {\scriptstyle f}\downarrow & \longmapsto & \uparrow{\scriptstyle f^*} \\ y & & \mathrm{Hom}_{\mathsf{C}}(y,c) \end{array}$$

▽

Definition 1.0.14. A Functor $F : \mathsf{C} \to \mathsf{Set}$ is **representable** if there exists $c \in \mathsf{C}$ such that F is naturally isomorphic to the represented functor associated to c. ⌟

Definition 1.0.15 (Limit)**.** ⌟

Definition 1.0.16 (Colimit)**.** ⌟

Definition 1.0.17 (Equalizer)**.** ⌟

Definition 1.0.18 (Adjoint)**.** ⌟

Definition 1.0.19 (Product)**.** ⌟

Definition 1.0.20 (Coproduct)**.** ⌟

Definition 1.0.21. ⌟

Definition 1.0.22 (Subcategory)**.** ⌟

Definition 1.0.23 (Full Subcategory)**.** ⌟

Definition 1.0.24 (Directed Colimit)**.** ⌟

Proposition 1.0.1 (Functors preserve Isomorphisms)**.**

Definition 1.0.25 (Universal property)**.** ⌟

Lemma 1.0.1 (Yonneda lemma)**.**

Definition 1.0.26 (Algebraic Category)**.** ⌟

Part II

Homological algebra

Chapter 2

Abelian categories

2.1 Additive categories

Definition 2.1.1 (Additive Category)**.** A Category A is **additive** if it verifies:

A.1) There exists a Zero object $0 \in \mathsf{A}$.

A.2) A has finite Products.

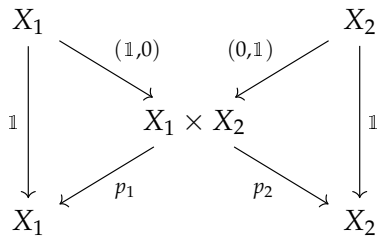
A.3) For every pair of objects $X, Y \in \mathsf{A}$ the set $\mathrm{Hom}_{\mathsf{A}}(X, Y)$ is endowed with an abelian group structure and the composition maps are bilinear with respect to the group operation.

⌟

Lemma 2.1.1. *Finite Products and Coproducts coincide in an Additive Category.*

Proof. Consider $X_1, X_2 \in \mathsf{A}$. We only need to check that $X_1 \times X_2 \cong X_1 \amalg X_2$ since the case for a finite collection of elements follows directly by induction. To do this, we will see that $X_1 \times X_2$ satisfies the universal property of the Coproduct.

Consider the following diagram



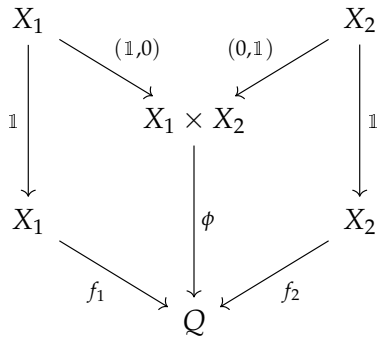
Note that for any pair of morphisms $(\alpha_1, \beta_1), (\alpha_2, \beta_2) : R \to X_1 \times X_2$ we have that

$$(\alpha_1, \beta_1) + (\alpha_2, \beta_2) = (\alpha_1 + \alpha_2, \beta_1, \beta_2)$$

Therefore if p_1, p_2 are two projections on $X_1 \times X_2$ whose sum is the identity we have that

$$(\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2 = \mathbb{1}_{X_1 \times X_2}$$

Consider now two morphisms $f_i : X_i \to Q$. Then its sum $\phi = f_1 \circ p_1 + f_2 \circ p_2$ produces the following diagram



Furthermore, this is unique. Indeed, consider $\phi' : X_1 \times X_2 \to Q$ another morphism filling the diagram. Then

$$\begin{aligned}
\phi &= \phi \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi \circ (id, 0) \circ p_1 + \phi \circ (0, \mathbb{1}) \circ p_2 \\
&= f_1 \circ p_1 + f_2 \circ p_2 \\
&= \phi' \circ (\mathbb{1}, 0) + \phi' \circ (0, \mathbb{1}) \\
&= \phi' \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi' \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi'
\end{aligned}$$

Therefore $X_1 \times X_2$ verifies the universal property of $X_1 \amalg X_2$. □

Notation. If $\{X_i\}_{i=1}^n$ is a finite set of objects in an Additive Category we will denote its Product (and therefore by Lemma 2.1.1 its Coproduct) by $\bigoplus_{i=1}^n X_i$.

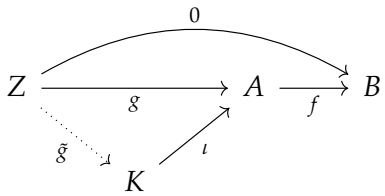
Lemma 2.1.2. *A morphism $\varphi : A \to B$ in an Additive Category is a monomorphism if and only if for every morphism $\xi : Z \to A$*

$$\varphi \circ \xi = 0 \implies \xi = 0$$

Dually φ is an epimorphism if and only if for every morphism $\xi : B \to Z$

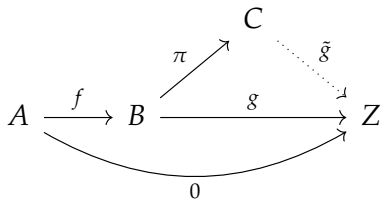
$$\xi \circ \varphi = 0 \implies \xi = 0$$

Definition 2.1.2 (Kernel)**.** Let $f : A \to B$ be a morphism in an Additive Category A. A morphism $\iota : K \to A$ is a **kernel** of f if $f \circ \iota = 0$ and for every morphism $g : Z \to A$ such that $f \circ g = 0$ there exists a unique $\tilde{g} : Z \to K$ making the following diagram commute



⌟

Definition 2.1.3 (Cokernel)**.** Let $f : A \to B$ be a morphism in an Additive Category A. A morphism $\pi : B \to C$ is a **cokernel** of f if $\pi \circ f = 0$ and for every morphism $g : B \to Z$ such that $\pi \circ g = 0$ there exists a unique $\tilde{g} : C \to Z$ making the following diagram commute

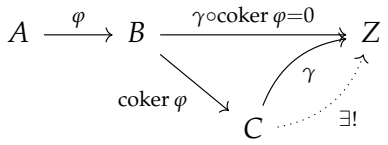


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Lemma 2.1.3. *In an Additive Category Kernels are monomorphisms and Cokernels are epimorphisms.*

Proof. We will only show that Cokernels are epimorphisms. The statement about Kernels follows analogously.

Let $\varphi : A \to B$ be a morphism and let coker $\varphi : B \to C$ be its Cokernel. Consider another morphism $\gamma : C \to Z$ such that $\gamma \circ \operatorname{coker} \varphi = 0$. Since $(\gamma \circ \operatorname{coker} \varphi) \circ \varphi = 0$ by definition of the Cokernel we have



Therefore since $\gamma \circ \operatorname{coker} \varphi = 0 = 0 \circ \operatorname{coker} \varphi$ uniqueness forces $\gamma = 0$ and thus by Lemma 2.1.2 φ is an epimorphism. □

Lemma 2.1.4. *Let $\varphi : A \to B$ be a morphism in an Additive Category. Then φ is a monomorphism if and only if $0 \to A$ is its Kernel and φ is an epimorphism if and only if $B \to 0$ is its Cokernel.*

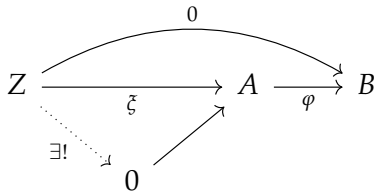
Proof. We will do the Kernel one.

$\Longrightarrow$ Suppose φ is a monomorphism.

If $\xi : Z \to A$ is a morphism such that $\varphi \circ \xi = 0$ then by Lemma 2.1.2 $\xi = 0$. In particular ξ factors uniquely through $0 \to A$, i.e: $0 \to A$ is the Kernel of φ.

$\Longleftarrow$ Suppose now that $0 \to A$ is the Kernel of φ.

Consider a morphism $\xi : Z \to A$ such that $\varphi \circ \xi = 0$. By the Kernel properties we have the following diagram



Therefore by uniqueness $\xi = 0$, making φ a monomorphism by Lemma 2.1.2.

□

Definition 2.1.4 (Additive Functor)**.** Let A, B be two Additive Categories. A Functor $F : \mathsf{A} \to \mathsf{B}$ is **additive** if it preserves the abelian group structure of all Hom-sets. ⌟

2.2 Abelian categories

Definition 2.2.1 (Abelian Category)**.** An Additive Category A is **abelian** if it verifies

A.4) Kernels and Cokernels exist in A.

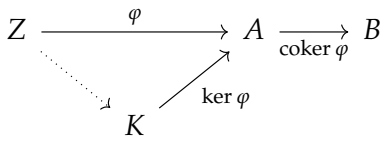
A.5) Every monomorphism is the Kernel of some morphism and every epimorphism is the Cokernel of some morphism.

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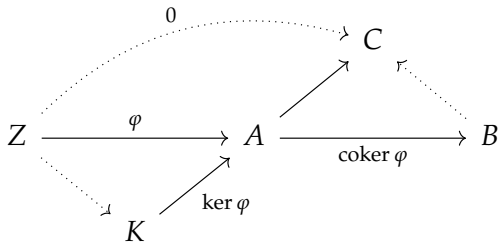
Lemma 2.2.1. *In an Abelian Category* A *every Kernel is the Kernel of its Cokernel and every Cokernel is the Cokernel of its Kernel.*

Proof. Lets check Cokernels this time.

Let coker $\varphi : A \to B$ be the Cokernel of a morphism $\varphi : Z \to A$. Since A is abelian φ has a kernel $\ker \varphi : K \to A$. Note that $Z \to A \to B$ is 0, therefore by the Kernel universal property we have



Consider a morphism $A \to C$ such that $(K \to A \to C) = 0$. Now using the Cokernel universal property we obtain

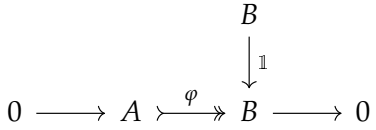


Therefore coker φ satisfies the universal property of the Cokernel of $\ker \varphi : K \to A$. □

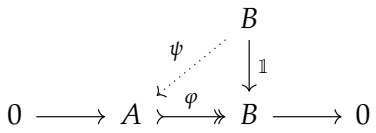
Lemma 2.2.2. *Let $\varphi : A \to B$ be a morphism in an Abelian Category* A*. If φ is both a monomorphism and epimorphism then φ is an Isomorphism.*

Proof. By Lemma 2.1.4 the Kernel of φ is $0 \to A$ and the Cokernel is $B \to 0$. Furthermore by Lemma 2.2.1 φ is the Cokernel of $0 \to A$ and the Kernel of $B \to 0$.

Consider the following diagram

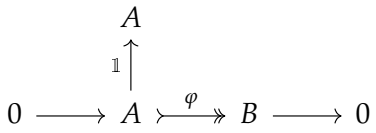


Since $(B \to B \to 0) = 0$ and φ is the Kernel of $B \to 0$ we have

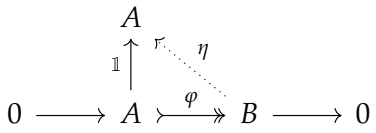


Therefore since $\varphi\psi = \mathbb{1}$ we have shown that φ has a right-inverse.

Consider now the diagram



Since $(0 \to A \to A) = 0$ and φ is the Cokernel of $0 \to A$ we have



Therefore since $\eta\varphi = \mathbb{1}$ we have shown that φ has a left-inverse. □

Definition 2.2.2 (Exact sequence)**.** A sequence of morphisms $X \xrightarrow{f} Y \xrightarrow{g} Z$ in an Abelian Category is **exact** (at Y) if $g \circ f = 0$ and the natural morphism $\operatorname{im} f \to \ker g$ is an Isomorphism.

More generally, a sequence of morphisms $\cdots \to X \to Y \to Z \to \cdots$ is **exact** if every successive pair of arrows is exact. ⌟

Remark 2.2.1. For any morphism $f : X \to Y$ in an Abelian Category we get exact sequences

$$0 \longrightarrow \ker f \longrightarrow X \longrightarrow \operatorname{im} f \longrightarrow 0$$

$$0 \longrightarrow \operatorname{im} f \longrightarrow Y \longrightarrow \operatorname{coker} f \longrightarrow 0$$

Therefore by Lemma 2.1.4 the sequence $0 \to X \xrightarrow{f} Y$ (resp. $X \xrightarrow{f} Y \to 0$) is exact if and only if f is a monomorphism (resp. epimorphism). ▽

Definition 2.2.3 (Exact functor)**.** Let A, B be two Abelian Categorys.

A Functor $F : \mathsf{A} \to \mathsf{A}'$ is called left (resp. right) **exact** if for any Exact sequence in A

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z$$

(resp. $X \to Y \to Z \to 0$) the squence

$$0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z)$$

(resp. $F(X) \to F(Y) \to F(Z) \to 0$) is exact.

If F is both left and right exact, then F is called **exact**. ⌟

Example 2.2.1. Let A be an Abelian Category and let $X \in \mathsf{A}$. The Functors $\operatorname{Hom}_{\mathsf{A}}(X, -)$ and $\operatorname{Hom}_{\mathsf{A}}(-, X)$ are both left-exact functors from A to Ab. ◊

Definition 2.2.4. Let A be an Abelian Category.

- An object $P \in \mathsf{A}$ is **projective** if the Functor $\mathrm{Hom}_{\mathsf{A}}(P, -)$ is exact.
- An object $I \in \mathsf{A}$ is **injective** if the Functor $\mathrm{Hom}_{\mathsf{A}}(-, I)$ is exact.

⌟

Definition 2.2.5. An Abelian Category A.

- A has **enough projectives** if for every object $A \in \mathsf{A}$ there exists a projective object $P \in \mathsf{A}$ and an epimorphism $P \twoheadrightarrow A$.
- A has **enough injectives** if for every object $A \in \mathsf{A}$ there exists a injective object $I \in \mathsf{A}$ and an monomorphism $A \rightarrowtail I$.

⌟

Theorem 2.2.1 (Freyd-Mitchell theorem)**.** *Let* A *be a small Abelian Category. Then there exists a fully faithfull Exact functor* $\mathsf{A} \to R\text{-}\mathsf{Mod}$ *for a suitable (possibly non-commutative) ring* R*, where* $R\text{-}\mathsf{Mod}$ *is the category of left* R*-modules.*

Chapter 3

Old

3.1 Complexes and (co)homology

From now on, we will be working in a fixed Abelian Category A. By the Freyd-Mitchell theorem we can work with objects and morphisms using set-theoretic tools when needed.

Definition 3.1.1 (Complex)**.** A sequence of objects and morphisms in an Additive Category A

$$\cdots \xrightarrow{d_{i-2}} C_{i-1} \xrightarrow{d_{i-1}} C_i \xrightarrow{d_i} C_{i+1} \xrightarrow{d_{i+1}} \cdots$$

is a **chain complex** if for every $i \in \mathbb{N} \quad d_i \circ d_{i-1} = 0$. We denote a chain complex as $(C_\bullet, d_\bullet)$ or simply as $C_\bullet$.

By reversing the order, we say that a sequence

$$\cdots \xrightarrow{d^{i+2}} C^{i+1} \xrightarrow{d^{i+1}} C^i \xrightarrow{d^i} C^{i-1} \xrightarrow{d^{i-1}} \cdots$$

is a **cochain complex** if for every $i \in \mathbb{N} \quad d_i \circ d_{i+1} = 0$. We denote a cochain complex as $(C^\bullet, d^\bullet)$ or simply as $C^\bullet$. ⌟

Remark 3.1.1. In purelly algebraic terms, chains are cochains are *equivalent*. We will adopt the cochain notation. ▽

Definition 3.1.2 (Exact complex)**.** A Complex in an Abelian Category

$$\cdots \xrightarrow{d^{i+2}} C^{i+1} \xrightarrow{d^{i+1}} C^i \xrightarrow{d^i} C^{i-1} \xrightarrow{d^{i-1}} \cdots$$

is **exact at** C^i (or simply at position i) if

$$\operatorname{coker} d^{i+1} \circ \ker d^i = 0$$

A complex is **exact** if it is exact at every position. ⌟

Definition 3.1.3 (Image and coimage)**.** Let $\varphi : A \to B$ be a morphism in an Abelian Category.

The **image** of φ is $\operatorname{im} \varphi = \ker(\operatorname{coker} \varphi)$ and the **coimage** of φ is $\operatorname{coim} \varphi = \ker(\operatorname{coker} \varphi)$. ⌟

Definition 3.1.4 (Cohomology)**.** Let $C^\bullet$ be a Complex in an Abelian Category A. We define the i**-th cohomology object** of $C^\bullet$ as

$$\mathrm{H}^i(C^\bullet) = {\ker d^i}/{\operatorname{im} d^{i-1}}$$

⌟

Definition 3.1.5 (Category of complexes)**.** Let A be an Abelian Category. The **Category of Complexes** $\mathsf{C}(\mathsf{A})$ consists of:

1. Cochain complexes in A as objects.
2. For $(M^\bullet, d^\bullet_{M^\bullet}), (N^\bullet, d^\bullet_{N^\bullet}) \in \mathsf{C}(\mathsf{A})$, $\mathrm{Hom}_{\mathsf{C}(\mathsf{A})}(M^\bullet, N^\bullet)$ consists of the commutative diagrams of the form

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & M^{i-1} & \xrightarrow{d^{i-1}_{M^\bullet}} & M^i & \xrightarrow{d^i_{M^\bullet}} & M^{i+1} & \xrightarrow{d^{i+1}_{M^\bullet}} & \cdots \\ & & \downarrow{\scriptstyle \alpha^{i-1}} & & \downarrow{\scriptstyle \alpha^i} & & \downarrow{\scriptstyle \alpha^{i+1}} & & \\ \cdots & \longrightarrow & N^{i-1} & \xrightarrow{d^{i-1}_{N^\bullet}} & N^i & \xrightarrow{d^i_{N^\bullet}} & N^{i+1} & \xrightarrow{d^{i+1}_{N^\bullet}} & \cdots \end{array}$$

   We denote by $\alpha^\bullet : M^\bullet \to N^\bullet$ the morphism determined by the collection of α^i.

⌟

Lemma 3.1.1. *The Category of complexes of an Abelian Category is an Abelian Category.*

Lemma 3.1.2. *Let A be a Abelian Category and let $\mathsf{C}(\mathsf{A})$ be its Category of complexes.*

Then for every integer i, the assignement $\mathrm{H}^i : M^\bullet \mapsto H^i(M^\bullet)$ taking a Complex $M^\bullet$ to its i-th Cohomology is a Functor $\mathsf{C}(\mathsf{A}) \to \mathsf{A}$.

Proof. We want to show that for all Complexes $M^\bullet, N^\bullet \in \mathsf{C}(\mathsf{A})$ and $i \in \mathbb{Z}$, H^i induces in a natural way a homomorphism of abelian groups

$$\mathrm{Hom}_{\mathsf{C}(\mathsf{A})}(M^\bullet, N^\bullet) \to \mathrm{Hom}_{\mathsf{A}}(\mathrm{H}^i(M^\bullet), \mathrm{H}^i(N^\bullet))$$

Let $\alpha^\bullet : M^\bullet \to N^\bullet$ be a morphism in $\mathsf{C}(\mathsf{A})$. Let's consider the relevant of the commutative diagram defining $\alpha^\bullet$:

$$\begin{array}{ccccc} M^{i-1} & \xrightarrow{d^{i-1}_{M^\bullet}} & M^i & \xrightarrow{d^i_{M^\bullet}} & M^{i+1} \\ \downarrow{\scriptstyle \alpha^{i-1}} & & \downarrow{\scriptstyle \alpha^i} & & \downarrow{\scriptstyle \alpha^{i+1}} \\ N^{i-1} & \xrightarrow{d^{i-1}_{N^\bullet}} & N^i & \xrightarrow{d^i_{N^\bullet}} & N^{i+1} \end{array}$$

By the commutativity of the right square we have that $d^i_{N^\bullet} \circ \alpha^i = \alpha^{i+1} \circ d^i_{M^\bullet}$ is 0 on $\ker d^i$. Therefore the restriction of α^i to $\ker d^i_{M^\bullet}$ factors through $\ker d^i_{N^\bullet}$. Composing with the projection we get a morphism

$$\ker d^i_{M^\bullet} \to \mathrm{H}^i(N^\bullet)$$

Since this morphism is 0 when restricted to $\mathrm{im}\, d^{i-1}_{M^\bullet}$ by the commutativity of the left square we get that α^i induces a morphism

$$\mathrm{H}^i(M^\bullet) \to \mathrm{H}^i(N^\bullet)$$

in A as we wanted. □

Remark 3.1.2. We can view Cohomology as a Functor $\mathsf{C}(\mathsf{A}) \to \mathsf{C}(\mathsf{A})$ by placing each Cohomology object $\mathrm{H}^i(M^\bullet)$ in degree i and connecting them using 0 morphisms, obtaining a complex $\mathrm{H}^\bullet(M^\bullet)$.

Note that the Cohomology of this particular Complex verifies

$$\mathrm{H}^\bullet(\mathrm{H}^\bullet(M^\bullet)) = \mathrm{H}^\bullet(M^\bullet)$$

∇

Part III

Sheaves

Chapter 4

Presheaves and sheaves

4.1 Presheaves

We begin by giving a general definition of a presheaf.

Definition 4.1.1 (Presheaf)**.** Let X be a topological space and let $\mathsf{Op}(X)$ be its Posetal category of open sets. Let C be a Category. The Category of C-valued **presheaves** is the `ex:functor-category`

$$\mathsf{PSh}_X(\mathsf{C}) := [\mathsf{Op}(X)^{\mathrm{op}}, \mathsf{C}]$$

When the topological space is clear we will write $\mathsf{PSh}(\mathsf{C})$.

The elements of this category are Functors $\mathcal{F} : \mathsf{Op}(X)^{\mathrm{op}} \to \mathsf{C}$ and morphisms are Natural transformations $\phi : \mathcal{F} \to \mathcal{G}$. ⌟

Let's rewrite Definition 4.1.1 less abstractly: a C-valued Presheaf $\mathcal{F}$ is the collection of data

- For every open set $U \subseteq X$ an object $\mathcal{F}(U)$ of C called *sections of $\mathcal{F}$ over U*.
- For every inclussion of open sets $V \subseteq U$ a morphism $\mathrm{res}_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V)$ in C called *restriction* such that
  - For every open set U, $\mathrm{res}_{U,U} = \mathbb{1}_U$.
  - For every inclussion of open sets $W \subseteq V \subseteq U$, $\mathrm{res}_{U,V} \circ \mathrm{res}_{V,W} = \mathrm{res}_{U,W}$.

A *morphism of Presheaves* $\phi : \mathcal{F} \to \mathcal{G}$ is a collection of morphisms $\{\phi(U) : \mathcal{F}(U) \to \mathcal{G}(U)\}_{U \in \mathsf{Op}(X)}$ in C making the following diagram commute:

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\phi(U)} & \mathcal{G}(U) \\ \downarrow{\scriptstyle \mathrm{res}_{U,V}} & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ \mathcal{F}(V) & \xrightarrow{\phi(V)} & \mathcal{G}(V) \end{array}$$

We are going to study now Limits and Colimits in $\mathsf{PSh}(\mathsf{C})$.

Note that, since $\mathsf{PSh}(\mathsf{C})$ is a `def:functor-category` then if C has Limits or Colimits of a certain shape then $\mathsf{PSh}(\mathsf{C})$ also has them and they are computed pointwise.

This means that Limits and Colimits in $\mathsf{PSh}(\mathsf{C})$ can be computed *open set by open set*.

Proposition 4.1.1. *Let X be a topological space. Let $\mathcal{I} \to \mathsf{PSh}_X(\mathsf{C}), i \mapsto \mathcal{F}_i$ be a Diagram. If C has Limits and/or Colimits with shape $\mathcal{I}$ then $\lim_i \mathcal{F}_i$ and/or $\mathrm{colim}_i \mathcal{F}_i$ exist in $\mathsf{PSh}_X(\mathsf{C})$.*

Moreover for every open set $U \subseteq X$

$$\left(\lim_i \mathcal{F}_i\right)(U) = \lim_i \mathcal{F}_i(U) \qquad \left(\operatorname*{colim}_i \mathcal{F}_i\right)(U) = \operatorname*{colim}_i \mathcal{F}_i(U)$$

Write a section on functor categories (or quote it from somewhere) to justify this.

In particular if C is an Abelian Category then Kernels and Cokernels exist and are computed *open set by open set*, making $\mathsf{PSh}(\mathsf{C})$ an Abelian Category.

Remark 4.1.1. From this point onwards, we will suppose C has all Limits and Colimits unless stated otherwise. ▽

4.1.1 Presheaves and continuous maps

Definition 4.1.2 (Direct image Presheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathcal{F}$ be a C-valued Presheaf over X. The **direct image Presheaf** of $\mathcal{F}$ via f is the Presheaf $f_*\mathcal{F}$ defined as, for every open subset $V \subseteq Y$

$$f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}(U))$$

Given $W \subseteq V \subseteq Y$ open subsets, the restriction map comes from the commutativity of the diagram

$$\begin{array}{ccc} f_*\mathcal{F}(V) & = & \mathcal{F}(f^{-1}V) \\ \vdots & & \downarrow \\ f_*\mathcal{F}(V) & = & \mathcal{F}(f^{-1}V) \end{array}$$

This construction defines a Functor $f_* : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{PSh}_Y(\mathsf{C})$. ⌟

Show that the category of presheaves over a one point space *is equal* to the evaluation category.

Example 4.1.1 (Global section)**.** Let X be a topological space. Consider the one-point topological space $\{*\}$. Let $\mathcal{F}$ be a C-valued presheaf over X. The **global section** of $\mathcal{F}$ over

This defines a Functor $\Gamma(X, -) : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{C}$ ◊

Example 4.1.2 (Skyscrapper Presheaf)**.** Let X be a topological space and $p \in X$. Suppose C has a final object $* \in \mathsf{C}$. Let S be an object in C. The **skyscrapper Presheaf** supported at S $\iota_{p,*}\underline{S}_{\text{pre}}$ is the Direct image Presheaf of the Constant Presheaf $\underline{S}_{\text{pre}}$ via the inclusion $\iota : \{p\} \hookrightarrow X$:

$$(\iota_{p,*}\underline{S}_{\text{pre}})(U) = \begin{cases} S & p \in U \\ * & p \notin U \end{cases}$$

◊

Using this we can construct a *dual* Presheaf to Definition 4.1.2.

Definition 4.1.3 (Inverse image Presheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathcal{G}$ be a C-valued Presheaf over Y. The **inverse image Presheaf** of $\mathcal{G}$ via f is the Presheaf $f^{-1}\mathcal{G}$ defined as, for every open subset $U \subseteq X$

$$f^{-1}\mathcal{G}(U) = \operatorname*{colim}_{V \supseteq f(U)} \mathcal{G}(V)$$

where the Colimit is taken over all the open neighbourhoods V of $f(U)$.

This defines a Functor $f^{-1} : \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$. ⌟

Remark 4.1.2. Note that the set of open neighbourhoods of $f(U)$ is a partially ordered set via reverse inclussion. Therefore if C is a concrete category we can describe $f^{-1}\mathcal{G}(U)$ as $\{(s, V) \mid f(U) \subseteq V \in \mathsf{Op}(Y), s \in \mathcal{G}(V)\}$ modulo the relation $(s, V) \sim (s', V')$ if and only if there exists $W \subseteq V \cap V'$ open with $f(U) \subseteq W$ such that $s|_W = s'|_W$. ▽

Example 4.1.3 (Constant Presheaf)**.** Let X be a topological space. Let S be an object of C. The **constant Presheaf** associated to S $\underline{S}_{\text{pre}}$ is the Inverse image Presheaf of S via the map $\{*\} \to X$. ◊

Definition 4.1.4 (Stalk)**.** Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued Presheaf over X. The **stalk** of $\mathcal{F}$ at p $\mathcal{F}_p$ is the Inverse image Presheaf of $\mathcal{F}$ via $X \to \{p\}$:

$$\mathcal{F}_p = \operatorname*{colim}_{U \ni p} \mathcal{F}(U)$$

⌟

Proposition 4.1.2. *Let $f : X \to Y$ be a continuous map. We have an* `def:adjunction` *between the Direct image Presheaf functor and the Inverse image Presheaf $f^{-1} \dashv f_*$.*

Proof. We define the `def:categorical-unit` $\eta : 1 \implies f_* f^{-1}$ as the map with components given by the canonical insertion maps to the colimit

$$\mathcal{F}(W) \to \operatorname*{colim}_{U \supseteq (f \circ f^{-1})(W)} \mathcal{F}(U) = f^{-1}\mathcal{F}(f^{-1}W) = f_* f^{-1}\mathcal{F}(W)$$

Note that this maps exist because $(f \circ f^{-1})(W) \subseteq W$.

On the other hand, note that if $f(V) \subseteq U$ then $V \subseteq (f^{-1} \circ f)(V) \subseteq f^{-1}(U)$, i.e: we have a restricion map $\mathcal{G}(f^{-1}(U)) \to \mathcal{G}(V)$. Therefore by the universal property of the colimit we have a unique map

$$f^{-1} f_* \mathcal{G}(V) = \operatorname*{colim}_{U \supseteq f(V)} \mathcal{G}(f^{-1}(U)) \to \mathcal{G}(V)$$

Using this maps we obtain a `def:categorical-counit` $\varepsilon : f^{-1} f_* \implies 1$.

Since by construction the compositions $\eta \circ \varepsilon$ and $\varepsilon \circ \eta$ are the corresponding identity morphisms, we have proven the adjunction. □

Two important particular cases of the adjunction Proposition 4.1.2 are

Corollary 4.1.1. *Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued Presheaves over X and let A be an object in C. Then*

$$\mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\underline{A}_{\text{pre}}, \mathcal{F}) \simeq \mathrm{Hom}_{\mathsf{C}}(A, \Gamma(X, \mathcal{F}))$$

Corollary 4.1.2. *Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued Presheaves over X and let A be an object in C. Then*

$$\mathrm{Hom}_{\mathsf{C}}(\mathcal{F}_p, A) \simeq \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota_{p,*}\underline{A}_{\text{pre}})$$

4.2 Sheaves

We now turn our attention to a particular class of Presheaves, following [**stacks-project**].

Let's start by considering the case $\mathsf{C} = \mathsf{Set}$:

Definition 4.2.1 (Sheaf of sets)**.** Let X be a topological space. A Presheaf of sets $\mathcal{F}$ is a **sheaf** if for every open set U and open covering $\{U_i\}_{i\in\mathcal{I}}$ of U the diagram

$$\mathcal{F}(U) \to \prod_{i\in\mathcal{I}} \mathcal{F}(U_i) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathcal{F}(U_i \cap U_j) \tag{4.2.1}$$

is an Equalizer.

We denote by $\mathsf{Sh}_X(\mathsf{Set})$ (or just Sh if the context is clear) the Full Subcategory of the category of presheaves of sets whose objects are sheaves of sets.

Less abstractly this means that $\mathcal{F}$ verifies:

SH.1) **Identity**: Given two sections $s, t \in \mathscr{F}(U)$ such that for every $i \in \mathcal{I}$ $\quad s|_{U_i} = t|_{U_i}$ we have that $s = t$.

SH.2) **Gluability**: Given a collection of sections $\{s_i \mid s_i \in \mathscr{F}(U_i)\}_{i\in\mathcal{I}}$ that agree on pairwise intersections ($\forall i, j \in \mathcal{I}\ \ s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$) there exists a section $s \in \mathscr{F}(U)$ such that $\forall i \in \mathcal{I}\ \ s|_{U_i} = s_i$.

⌟

Definition 4.2.2 (Category of Sheaves of sets)**.** The Category $\mathsf{Sh}_X(\mathsf{Set})$ of Sheaves of sets is the Full Subcategory of the category of Presheaves $\mathsf{PSh}_X(\mathsf{Set})$ whose objects are Sheaves of sets. ⌟

Let's try to make Definition 4.2.1 more general by considering another evaluation category C.

Firstly, note that if Products indexed by $\mathcal{I}$ and $\mathcal{I}\times\mathcal{I}$ for any indexing set $\mathcal{I}$ exist in C then the diagram (4.2.1) makes sense. Secondly, we have that (4.2.1) is an Equalizer if and only if for every object $C \in \mathsf{C}$ the diagram in Set

$$\mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U)) \to \prod_{i\in\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U_i)) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U_i \cap U_j))$$

is an Equalizer.

Therefore we get

Definition 4.2.3 (Sheaf)**.** Let X be a topological space. Let C be an arbitrary category. A C-valued Presheaf $\mathcal{F}$ is a **sheaf** if for all objects $C \in \mathsf{C}$ the Presheaf of sets $\mathcal{F}_C$ defined as $\mathcal{F}_C(U) = \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U))$ is a Sheaf of sets.

We denote by $\mathsf{Sh}_X(\mathsf{C})$ the Full Subcategory of the category of C-valued presheaves of sets whose objects are C-valued sheaves. ⌟

Notation. From now on, to distinguish a Presheaf from a Sheaf we will denote the former by $\mathcal{F}, \mathcal{G}, \mathcal{H}, \ldots$ and the latter by $\mathscr{F}, \mathscr{G}, \mathscr{H}, \ldots$

By Definition 4.2.3 we have an inclussion functor $\iota : \mathsf{Sh}(\mathsf{C}) \hookrightarrow \mathsf{PSh}(\mathsf{C})$.

For *sufficiently nice* Categories, this Functor admits a left-adjoint $\mathsf{PSh}(\mathsf{C}) \to \mathsf{Sh}(\mathsf{C})$. Let's study first the case $\mathsf{C} = \mathsf{Set}$.

How general do we want to go?

Theorem 4.2.1 (Sheafification). *There exists a* ***sheafification*** *Functor* $(-)^{\#} : \mathsf{PSh}(\mathsf{Set}) \to \mathsf{Sh}(\mathsf{Set})$ *that is left-adjoint to the inclussion Functor* $\iota : \mathsf{Sh}(\mathsf{Set}) \to \mathsf{PSh}(\mathsf{Set})$.

Proof. Let $\mathcal{F}$ be a Presheaf of sets over X. For every open subset $U \subseteq X$ we define

$$\mathcal{F}^{\#}(U) = \left\{ f_p \in \prod_{p \in U} \mathcal{F}_p \text{ verifying } (*) \right\}$$

where $(*)$ is the property *"for all $p \in U$ there exists an open neighbourhood $V \subseteq U$ of p and a section $s \in \mathcal{F}(V)$ such that, for every $q \in V$, $s_q = f_q$"*.

Taking $\prod_{p \in U} \mathcal{F}_p \to \prod_{q \in V} \mathcal{F}_q$ as the restriction maps we see that $\mathcal{F}^{\#}$ is a Presheaf. Let's see now that $\mathcal{F}^{\#}$ is a Sheaf. To do this, consider an open subset $U \subseteq X$ with open cover $\{U_i\}_{i \in \mathcal{I}}$.

SH.1 Let $s, t \in \mathcal{F}^{\#}(U)$ be two sections such that, for all $i \in \mathcal{I}$ $s|_{U_i} = t|_{U_i}$.

Then since for every $p \in U$ there exists $i \in \mathcal{I}$ such that $p \in U_i$ we have that $s_p = (s|_{U_i})_p = (t|_{u_i})_p = t_p$. Therefore $s = t$.

SH.2 Let $\{s_i \in \mathcal{F}^{\#}(U_i)\}_{i \in \mathcal{I}}$ be a collection of sections such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in \mathcal{I}$.

Consider $s \in \prod_{p \in U} \mathcal{F}_p$ defined as $s_p = (s_i)_p$ if $p \in U_i$. Note that this is well-defined by property $(*)$. To see that s verifies $(*)$ note that for every $p \in U_i$ there exists an open neighbourhood $V \subseteq U$ of p and $t \in \mathcal{F}(V)$ such that, for all $q \in V$, $t_q = (s_i)|_q = s_q$.

Let $V \subseteq U \subseteq X$ be two open sets. Consider the commutative diagram

$$\begin{array}{ccccc} \mathcal{F}(U) & \longrightarrow & \mathcal{F}^{\#}(U) & \longrightarrow & \prod_{p \in U} \mathcal{F}_p \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{F}(V) & \longrightarrow & \mathcal{F}^{\#}(V) & \longrightarrow & \prod_{q \in V} \mathcal{F}_q \end{array}$$

where the vertical arrows are the corresponding restriction maps. By chasing this diagram we get a map $\varphi : \mathcal{F} \to \mathcal{F}^{\#}$, $\varphi_U(s) = (s_p)_{p \in U}$.

It only remains to show that any morphism $\psi : \mathcal{F} \to \mathcal{G}$, where $\mathcal{G}$ is a Sheaf of sets, factors uniquely as $\mathcal{F} \xrightarrow{\varphi} \mathcal{F}^{\#} \to \mathcal{G}$, i.e: we have to construct a morphism $\theta : \mathcal{F}^{\#} \to \mathcal{G}$ such that $\psi = \theta \circ \varphi$.

Let $U \subseteq X$ be an open subset and $s \in \mathcal{F}^{\#}(U)$. For each $p \in U$ consider an open neighbourhood $V^{(p)} \subseteq U$ of p and $t^{(p)} \in \mathcal{F}(V^{(p)})$ such that $t^{(p)}_q = s_q$. Consider now $\psi_{V^{(p)}}(t^{(p)}) \in \mathcal{G}(V^{(p)})$. Note that $\{\psi_{V^{(p)}}(t^{(p)})\}_{p \in U}$ is a compatible family of section in $\mathcal{G}$, therefore by the Sheaf axioms there exists a unique section $t \in \mathcal{G}(U)$ such that $t|_{V^{(p)}} = \psi_{V^{(p)}}(s^{(p)})$. Therefore we can define $\theta_U(s) = t$.

By construction $\psi = \theta \circ \varphi$ and θ commutes with the restriction maps, making it a morphism of sheaves.

To check uniqueness, let $\theta' : \mathcal{F}^{\#} \to \mathcal{G}$ be another morphism such that $\psi = \theta' \circ \varphi$. Consider $s \in \mathcal{F}^{\#}(U)$. For each $p \in U$ choose $V^{(p)} \subseteq U$ and $s^{(p)} \in \mathcal{F}(V^{(p)})$ such that $s|_{V^{(p)}} = \varphi_{V^{(p)}}(s^{(p)})$. Then

$$\theta'(s)|_{V^{(p)}} = \theta'(s|_{V^{(p)}}) = \theta'(\varphi_{V^{(p)}}(s^{(p)})) = \psi_{V^{(p)}}(s^{(p)})$$

On the other hand $\theta(s)|_{V^{(p)}} = \psi_{V^{(p)}}(s^{(p)})$ by definition. Therefore since $\mathscr{G}$ is a Sheaf by SH.1 we have $\theta'(f) = \theta(f)$, i.e: $\theta' = \theta$. □

We can extend the construction outlined in Theorem 4.2.1 to Algebraic categories.

Lemma 4.2.1. *Let (C, F) be an Algebraic Category. Let $\mathcal{F}$ be a C-valued Presheaf. Then there exists a C-valued Sheaf $\mathcal{F}^{\#}$ and a morphism of presheaves $\mathcal{F} \to \mathcal{F}^{\#}$ satisfying*

1. *The map $\mathcal{F} \to \mathcal{F}^{\#}$ identifies the underlying Sheaf of sets of $\mathcal{F}^{\#}$ with the sheafification of the underlying Presheaf of sets of $\mathcal{F}$.*
2. *For any morphism of presheaves $\mathcal{F} \to \mathscr{G}$ where $\mathscr{G}$ is a C-valued Sheaf there exists a unique factorization $\mathcal{F} \to \mathcal{F}^{\#} \to \mathscr{G}$.*

Remark 4.2.1. We will assume that C is an Algebraic Category in what follows. ▽

Therefore we can assign a Sheaf to any Presheaf using this construction.

Definition 4.2.4 (Sheafification)**.** Let $\mathcal{F}$ be a C-valued Presheaf. We call the Sheaf $\mathcal{F}^{\#}$ obtained by Theorem 4.2.1 the **sheafification** of $\mathcal{F}$. ⌟

Using this adjunction we can *adapt* Definition 4.1.2 and Definition 4.1.3 to the case of Sheaves

Definition 4.2.5 (Direct image Sheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathscr{F}$ be a C-valued Sheaf over X. The **direct image Sheaf** of $\mathscr{F}$ via f is the Sheaf $f_*\mathscr{F}$ defined as, for every open subset $V \subseteq Y$

$$f_*\mathscr{F}(V) = \mathscr{F}(f^{-1}(U))$$

This construction defines a Functor $f_* : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{Sh}_Y(\mathsf{C})$. ⌟

This means that Example 4.1.1 and Example 4.1.2 translate directly to Sheaves.

Definition 4.2.6 (Global section Functor)**.** We denote by $\Gamma(X, -) : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$ the **global section functor** described in Example 4.1.1, but now with initial category $\mathsf{Sh}_X(\mathsf{C})$. ⌟

Definition 4.2.7 (Skyscrapper Sheaf)**.** Let X be a topological space and $p \in X$. Let S be an object of C. The **skyscrapper Sheaf** supported at p $\iota_{*,p}\underline{S}$ is the Direct image Sheaf of the Constant Sheaf $\underline{S}$ via the map $\{p\} \to X$. ⌟

Rewrite this definitions

Explain why we need to sheafificate

Definition 4.2.8 (Inverse image Sheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathscr{G}$ be a C-valued Sheaf over Y. The **inverse image Sheaf** of $\mathscr{G}$ via f is the Sheaf $f^{-1}\mathscr{G}$ defined as, for every open subset $U \subseteq X$

$$f^{-1}\mathscr{G} = (f^{-1}\mathscr{G})^{\#}$$

where the RHS of the equation is the Sheafification of the Inverse image Presheaf of $\mathscr{G}$ via f.

This defines a Functor $f^{-1} : \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$. ⌟

Definition 4.2.9 (Constant Sheaf)**.** Let S be an object of C. The **constant Sheaf** $\underline{S}$ associated to S is the Sheafification of the Constant Presheaf $\underline{S}_{\text{pre}}$, i.e: $\underline{S} = (\underline{S}_{\text{pre}})^{\#}$. ⌟

Definition 4.2.10 (Stalk)**.** Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued Sheaf over X. The **stalk** of $\mathscr{F}$ at p $\mathscr{F}_p$ is the Inverse image Sheaf of $\mathscr{F}$ via $X \to \{p\}$:

$$\mathscr{F}_p = \operatorname*{colim}_{U \ni p} \mathscr{F}(U)$$

⌟

Using the adjointness we have the analogous of Proposition 4.1.2

Proposition 4.2.1. *Let $f : X \to Y$ be a continuous map. We have an* `def:adjunction` *between the Direct image Sheaf functor and the Inverse image Sheaf $f^{-1} \dashv f_*$.*

Proof.

$$\begin{aligned}\operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(f^{-1}\mathscr{G}, \mathscr{F}) &= \operatorname{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(f^{-1}\mathscr{G}, \mathscr{F}) \\ &= \operatorname{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathscr{G}, f_*\mathscr{G}) \\ &= \operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathscr{G}, f_*\mathscr{G})\end{aligned}$$

□

Therefore adjunctions Corollary 4.1.1 and Corollary 4.2.2 also happen when considering Sheaves.

Corollary 4.2.1. *Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued Sheaves over X and let A be an object in C. Then*

$$\operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\underline{A}, \mathscr{F}) \simeq \operatorname{Hom}_{\mathsf{C}}(A, \Gamma(X, \mathscr{F}))$$

Corollary 4.2.2. *Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued Sheaves over X and let A be an object in C. Then*

$$\operatorname{Hom}_{\mathsf{C}}(\mathscr{F}_p, A) \simeq \operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathscr{F}, \iota_{p,*}\underline{A})$$

For cohomology we will be interested in the case $\mathsf{C} = \mathsf{Ab}$.

Lemma 4.2.2. *Let $a : \mathsf{A} \to \mathsf{B}$ and $b : \mathsf{B} \to \mathsf{A}$ be two Functors. Suppose that*

1. *A and B are additive categories, a, b are additive functors and $b \dashv a$.*
2. *B is an Abelian Category and b is left-exact.*
3. *$ba \cong \mathbb{1}_{\mathsf{A}}$.*

Then A is abelian.

Proof. Since B is an Abelian Category all Limits and Colimits exist en B.

Since b is left-adjoint it is right-exact and therefore, exact.

Let $\varphi : B_1 \to B_2$ be a morphism in B. If $K = \ker(\varphi)$ then K is the Equalizer of the diagram

$$K \longrightarrow B_1 \underset{0}{\overset{\varphi}{\rightrightarrows}} B_2$$

therefore applying b we get

$$bK \longrightarrow bB_1 \underset{0}{\overset{b\varphi}{\rightrightarrows}} bB_2$$

i.e: bK is the Kernel of $b\varphi$.

Dually we get that $b\operatorname{coker}(B_1 \to B_2)$ is the Cokernel of $b\varphi$.

Therefore we have that every morphism $b\varphi$ in A admits a Kernel and a Cokernel. Moreover since $ba \cong \mathbb{1}_{\mathsf{A}}$ every morphism in A is of the form $b\varphi$, therefore all Kernels and Cokernels exist in A. Indeed for every $\psi : A_1 \to A_2$ we have that

$$\ker \psi = b\ker(a\psi) \qquad \operatorname{coker} \psi = b\operatorname{coker}(a\psi) \tag{4.2.2}$$

It only remains to show that $\operatorname{coim} \psi = \operatorname{im} \psi$. Since A has all Limits and Colimits A has all finite Limits and Colimits, making a left-exact. This means that a transforms Kernels into Kernels. Therefore

$$\begin{aligned}
\operatorname{coim} \psi &= \operatorname{coker}(\ker \psi \to A_1) && \text{(definition)} \\
&= b\operatorname{coker}(a(\ker \psi \to A_1)) && \text{(4.2.2)} \\
&= b\operatorname{coker}(\ker(a\psi) \to aA_1) && (a \text{ preserves kernels}) \\
&= b\operatorname{coim}(a\psi) && \text{(definition)} \\
&= b\operatorname{im}(a\psi) && (\mathsf{B} \text{ is abelian}) \\
&= b\ker(aA_2 \to \operatorname{coker}(a\psi)) && \text{(definition)} \\
&= \ker(baA_2 \to b\operatorname{coker}(a\psi)) && (b \text{ preserves kernels}) \\
&= \ker(A_2 \to b\operatorname{coker}(a\psi)) && ba \cong \mathbb{1}_{\mathsf{A}} \\
&= \ker(A_2 \to \operatorname{coker} \psi) && \text{(4.2.2)} \\
&= \operatorname{im} \psi && \text{(definition)}
\end{aligned}$$

□

Proposition 4.2.2. *Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism in $\mathsf{Sh}(\mathsf{Ab})$.*

1. *The Category $\mathsf{Sh}_X(\mathsf{Ab})$ is an Abelian Category.*
2. *The Kernel of φ is the same as the Kernel of φ in $\mathsf{PSh}(\mathsf{Ab})$.*
3. *φ is injective if and only if φ is injective as a map or presheaves if and only if φ is injective as a map of sheaves.*
4. *The Cokernel of φ is the Sheafification of the Cokernel of φ in $\mathsf{PSh}(\mathsf{Ab})$.*
5. *φ is surjective if and only if φ is surjective as a map of sheaves.*
6. *A complex $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact at $\mathscr{G}$ if and only if for all open sets U and sections $s \in \mathscr{G}(U)$ mapping to 0 in $\mathscr{H}(U)$ there exists an open covering $\{U_i\}_{i\in\mathcal{I}}$ such that each $s|_{U_i}$ is the image of $\mathscr{F}(U_i) \to \mathscr{G}(U_i)$.*

Proof. Note that 2. $\implies$ 3., 4. $\implies$ 5 and 2. $\wedge$ 5. $\implies$ 6.

Therefore we only need to prove 1., 2. and 3., but this comes directly from Lemma 4.2.2 with $\mathsf{A} = \mathsf{Sh}_X(\mathsf{Ab})$, $\mathsf{B} = \mathsf{PSh}_X(\mathsf{Ab})$, $a : \mathsf{Sh}_X(\mathsf{Ab}) \hookrightarrow \mathsf{PSh}_X(\mathsf{Ab})$ the inclussion Functor and $b : \mathsf{PSh}_X(\mathsf{Ab}) \to \mathsf{Sh}_X(\mathsf{Ab})$ the Sheafification Functor. □

Chapter 5

Ringed spaces and $\mathscr{O}_X$-modules.

5.1 Ringed spaces

Definition 5.1.1 (Ringed space)**.** A **ringed space** is a tuple $(X, \mathscr{O}_X)$ where X is a topological space and $\mathscr{O}_X$ is a Sheaf of rings over X.

We call $\mathscr{O}_X$ the *structure sheaf* of the ringed space $(X, \mathscr{O}_X)$. Sections of $\mathscr{O}_X$ over an open subset $U \subseteq X$ are called *functions on* U and functions on X are called (global) functions. ⌟

Example 5.1.1. Let and X be a topological space. Then the $\mathbb{R}$-Algebra of continuous functions $\mathcal{C}^0_X(U) = \{f : U \to \mathbb{R} \mid f \text{ continuous}\}$ with the usual restriction maps forms a Sheaf of rings over X. Therefore $(X, \mathcal{C}^0_X)$ is a Ringed space. ◊

Example 5.1.2. If for every point $p \in X$ the Stalks $\mathscr{O}_{X,p}$ of the structure sheaf are Local rings then we call $(X, \mathscr{O}_X)$ a **locally ringed space**. ◊

Definition 5.1.2 (Morphism of Ringed spaces)**.** Let $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ be two Ringed spaces. A **morphism of ringed spaces** $(X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a pair $(f, f^\sharp)$ where $f : X \to Y$ is a continuous function (i.e: a morphism in Top) and $f^\sharp : \mathscr{O}_Y \to f_*\mathscr{O}_X$ is a morphism of sheaves of rings (i.e: a morphism in $\mathrm{Sh}_{\mathrm{Ring}}(X)$), where $f_*\mathscr{O}_X$ is the Direct image Sheaf. ⌟

Maybe it is better to define $f^\sharp$ as an f-map. **[stacks-project]**

Example 5.1.3. Let $f : X \to Y$ be a continuous function. Consider the Ringed spaces of continuous functions $(X, \mathcal{C}^0_X)$, $(Y, \mathcal{C}^0_Y)$.

Then the Morphism of Ringed spaces induced by f can be defined for every $V \subseteq Y$ open as $f^\sharp(V) : \mathcal{C}^0_Y(V) \to \mathcal{C}^0_X(f^{-1}(V))$, $f^\sharp(V)(h) = h \circ f|_{f^{-1}V}$. ◊

5.2 $\mathscr{O}_X$ modules

Definition 5.2.1 ($\mathscr{O}_X$-module)**.** Let $(X, \mathscr{O}_X)$ be a Ringed space. A **Sheaf of $\mathscr{O}_X$ modules** (or simply an $\mathscr{O}_X$-module) is a Sheaf $\mathscr{F}$ such that for every open set $U \subseteq X$ $\mathscr{F}(U)$ is an $\mathscr{O}_X(U)$-Module compatible with the restriction maps, i.e: for every $f \in \mathscr{O}_X(U)$ and $s \in \mathscr{F}(U)$: $(f \cdot s)|_V = f|_V \cdot s|_V$. ⌟

Example 5.2.1. ... ◊

Definition 5.2.2 (Morphism of $\mathscr{O}_X$-modules)**.** Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules. A morphism $\phi : \mathscr{F} \to \mathscr{G}$ is a sheaf morphism that commutes with the multiplication of the structure sheaf $\mathscr{O}_X$. ⌟

Example 5.2.2. ... ◊

Therefore for every Ringed space $(X, \mathscr{O}_X)$ the collection of $\mathscr{O}_X$-modules forms a Category $\mathscr{O}_X$-Mod.

5.2.1 $\mathscr{O}_X$-Mod and sheaves of abelian groups.

Just as abelian groups are $\mathbb{Z}$-modules, Sheaves of abelian groups can be seen as $\underline{\mathbb{Z}}_X$-modules, where $\underline{\mathbb{Z}}_X$ is the `ex:constant-sheaf` with coefficients in the integers.

To show this, for every Sheaf of abelian groups $\mathscr{F}$ over X and every open set $U \subseteq X$ we have to define an action $\cdot : \underline{\mathbb{Z}}_X(U) \times \mathscr{F}(U) \to \mathscr{F}(U)$ compatible with the restriction maps.

Let $f \in \underline{\mathbb{Z}}_X(U)$ and $s \in \mathscr{F}(U)$. Since f is locally constant there exists an open cover $\{U_i\}_{i\in\mathcal{I}}$ of U such that for every $i \in \mathcal{I} \quad f|_{U_i} = n_i \in \mathbb{Z}$.

For every U_i consider $t_i = n_i \cdot s|_{U_i}$, where $n_i \cdot s|_{U_i}$ is shorthand for $s|_{U_i}$ added to itself n_i times. Note that this is well-defined because $\mathscr{F}(U_i)$ is an abelian group.

Let $U_i, U_j \in \{U_i\}_{i\in\mathcal{I}}$ such that $U_i \cap U_j = \varnothing$. For every $x \in U_i \cap U_j$ there exists an open neighbourhood $V \subseteq U_i \cap U_j$ of x were f is constant.

Therefore $n_i = n_j$ and $t_i|_{U_i\cap U_j} = t_j|_{U_i\cap U_j}$ and, applying axiom SH.2, we can define $f \cdot s = t$.

Checking the axioms in small enough open sets such that f is constant verifies that this action makes $\mathscr{F}$ an $\underline{\mathbb{Z}}_X$-module.

Do we have to actually write the check?

5.3 $\mathscr{O}_X$-Mod is abelian.

Let's check first that $\mathscr{O}_X$-Mod is a Additive Category.

Lemma 5.3.1. *The category $\mathscr{O}_X$-Mod admits a zero object.*

Proof. Consider the $\mathscr{O}_X$ module that sends every open set to $\{0\}$. □

Lemma 5.3.2. *The category $\mathscr{O}_X$-Mod has all binary biproducts. We define their direct sum a $\mathscr{F} \oplus \mathscr{G} = \mathscr{F} \times \mathscr{G}$.*

Proof. Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules. □

Theorem 5.3.1. *Let $\varphi : \mathscr{F} \to \mathscr{G}$ a Morphism of $\mathscr{O}_X$-modules. Then the kernel of φ is the subsheaf of $\mathscr{F}$ with sections*

$$(\ker\varphi)(U) = \{s \in \mathscr{F}(U) \mid \varphi(s) = 0\}$$

Furthermore for every $x \in X \quad (\ker\varphi)_x = \ker\varphi_x$.

Proof. If $V \subseteq U$

$$\begin{array}{ccccccc} 0 & \longrightarrow & (\ker\varphi)(U) & \longrightarrow & \mathscr{F}(U) & \xrightarrow{\varphi(U)} & \mathscr{G}(U) \\ & & \vdots & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ 0 & \longrightarrow & (\ker\varphi)(V) & \longrightarrow & \mathscr{F}(V) & \xrightarrow{\varphi(V)} & \mathscr{G}(V) \end{array}$$

□

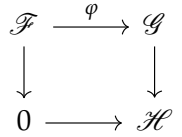
Theorem 5.3.2. *Let $\varphi : \mathscr{F} \to \mathscr{G}$ a Morphism of $\mathscr{O}_X$-modules. Then the cokernel of φ is the Sheafification of the quotient Presheaf*

$$(\operatorname{coker}_{\text{pre}} \varphi)(U) = {}^{\mathscr{G}(U)}\!/_{\varphi(\mathscr{F}(U))}$$

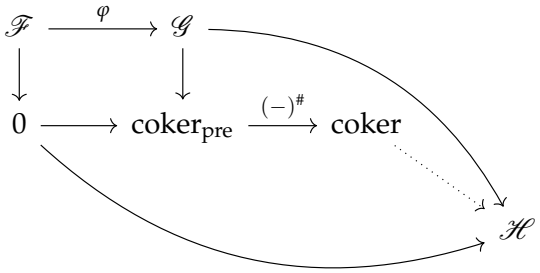
Furthermore for every $x \in X$ $\quad (\operatorname{coker} \varphi)_x = \operatorname{coker} \varphi_x$.

Prove that this is the cokernel in PSh.

Proof. Given a commutative diagram



We construct



□

Therefore we have proven that $\mathscr{O}_X$-Mod is an Abelian Category and in particular by $\mathscr{O}_X$-Mod and sheaves of abelian groups $\mathsf{Sh}_{\mathsf{Ab}}(X)$ is an Abelian Category.

5.4 Sequences in $\mathscr{O}_X$-Mod

Theorem 5.4.1. *The sequence of $\mathscr{O}_X$-modules $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact at $\mathscr{G}$ if and only if for every $x \in X$ the sequence $\mathscr{F}_x \to \mathscr{G}_x \to \mathscr{H}_x$ is exact at $\mathscr{G}_x$.*

Probably not a *theorem* but a direct consequence of the commutatitivity of stalks and (co)kernels

Maybe talk about resolutions / types of resolutions here?

Appendix A

Algebra

Definition A.0.1 (Algebra)**.** ⌟

Definition A.0.2 (Local ring)**.** ⌟

Definition A.0.3 (Module)**.** ⌟

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